

Analysis and Comparison of Road Freight Transport Cost in 20 European Countries

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ABSTRACT

Road transport is one of the most popular and practical means of freight transport in the world today. However, the cost of road freight transport may differ from one country to the next due to variances in a number of cost factors, including fuels, wages, taxation, tolls, insurance, maintenance, tyres, repairs, parking spaces, etc. The goal of this research was to measure and compare the cost of road freight transport in the 20 European countries with the highest recorded volume of tonne-kilometres, in order to draw conclusions about the cost competitiveness of road freight transport among them. Cost competitiveness in the sample was measured by 4 main cost categories: fuels, drivers' wages, tyres, and tolls. The results show large cost variations between countries. The countries found to have the lowest road freight transport cost were Lithuania, Poland and Bulgaria, and those with the highest costs were Norway, Austria, and the UK. The largest differences in costs were met in tolls and other road taxes, followed by drivers' wages, fuels, and finally tyres.

KEYWORDS

Analysis, Cost, Europe, Freight, Road, Transport

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1. INTRODUCTION

Road transport is one of the most popular and practical means of freight transport in the world today. However, the cost of road freight transport may differ from one country to the next due to variances in a number of cost factors, including fuels, wages, taxation, tolls, insurance, maintenance, repairs, tyres, parking space etc. The goal of this research is to measure and compare the cost of road freight transport, in the 20 European countries with the highest recorded volume of tonne-kilometres, in order to draw conclusions about the cost competitiveness of road freight transport among them. Cost competitiveness is measured by 4 main cost categories: fuels, drivers' wages, tyres, and tolls. The research starts with a short literature review about the history of road transport, an analysis of the categories and the quantity of road transport in Europe and an analysis of previous researches on the topic of road transport cost comparisons among European countries. Next is a description of the research's methodology, followed by the results, their analysis and the conclusions of the research.

2. LITERATURE REVIEW

2.1. Road Transport History

Road transport has developed alongside the development and evolution of humans. The first steps of humans were made in order to seek food, discover the natural environment and for protection from various natural hazards. However, because human forces are limited by nature, man has, through the use of various transportation inventions, managed to achieve a much faster and more effective transfer. The first and most important invention of man, related to the development of road transport, is the discovery of the wheel, 3,500 years ago in Mesopotamia (Britannica, 2018). By using the wheel in combination with the power of domesticated animals (horses, oxen, mules, etc.), he was able to carry large quantities of goods. Furthermore, in order to facilitate the transportation of wagons, he has built roads and rails (Freitag, 1979).

As regards road transport, its historical references have been recorded many centuries ago. The requirements for the movement and distribution of goods by large-scale road transport were mainly used at war campaigns. Great military leaders of history, such as Alexander the Great, Genghis Khan and Napoleon, relied on the design and implementation of road transport for maintaining their campaigns. Other important periods for the development of road transport were the British and American industrial revolutions which supported the expansion or construction of new trade routes, as well as the first and second world wars, which greatly increased the need for logistics in order to meet the demands of the combat forces (Erb & James, 2017). According to Dell, Moseley & Rand (2014) one of the most important reasons for the development of transport was also the discovery of the steam engine.

In the 20th century, the development of road freight transport was favored further. This growing supremacy stemmed both from the economic growth of cities and from the development of private businesses. Especially since the 1970s, significant investments were made in the transport sector. These investments were related to the construction of motorways, better vehicle technology and the use of computer programs and databases in order to gather useful information to support road transport services (Banister & Berechman, 1999). Completing the reasons for the development of road transport in the modern world, one should also mention globalization. Globalization is the process of interaction and integration between people, companies and governments worldwide and has grown due to advances in transportation and communication technology. Companies try to benefit from globalization in order to find the best markets for their products and services while securing the lowest cost resources. The enormous benefits that a company can gain through globalization have played an important role in the development of logistics and road transport in particular (Branch, 2009; Harrison & Hoek, 2012).

2.2. Road Transport in Europe

The transport sector is of critical importance for the economic development of European Union and also fulfils one the fundamental principles of the Community: the free movement of goods (Giannatos & Andrianopoulos, 1999). Freight services are divided into several categories and subcategories like sea, air, and land transport. Land transport is inextricably linked to land use, and its main advantages are related with accessibility and cost (Spiekermann & Urban, 2013). Land transport mainly involves rail and road transport. Road transport can be divided into national or domestic and international transport. The first category is the national road carriage of goods by the use of a commercial truck from Destination A to Destination B. The second category is the carriage of goods by truck from a country of origin and to another destination country (Theodoropoulou & Casole, 2014). Road freight transport is the most widespread type of transport because of its accessibility and its flexibility. The road network was the first way of transport created by man and allows access to more industrial sites than any other mean of transport. Road transport is also used to a significant extent, if not exclusively, in combined transport, such as in cases where the goods are transported either by sea, by river, by air or by rail or to any other corresponding intermediate infrastructure (Grant, Trautrimis & Yew Wong, 2015).

The European Union classifies vehicles as part of emission standards and other vehicle regulations. Passenger cars receive an “M” categorization, while commercial vehicles receive an “N” categorization (Directives 2002/24/EC of 18 March 2002 and 2007/46/EC of 5 September 2007). Category N vehicles are used for the carriage of goods and are divided in the following categories: a) N1 Vehicles for the carriage of goods and having a maximum mass not exceeding 3.5 tonnes, b) N2 Vehicles for the carriage of goods and having a maximum mass exceeding 3.5 tonnes but not exceeding 12 tonnes and c) N3 Vehicles for the carriage of goods and having a maximum mass exceeding 12 tonnes.

Road transport includes a large number of different types of freight transport trucks, covering all transport needs (transport of solid, liquid or gas cargo). However, there are two main categories of freight transport which are met on international markets: trucks without a drag (with refrigeration or not) and trucks with a drag (with refrigeration or not). These categories may offer the same storage space, but have different advantages and disadvantages. In the first case, trucks without a sliding load - superfluous, are suitable for long distance transport and with as few as possible recipients. The second solution, trucks with towed loads - trailers or slings, is suitable for a combination of many and few recipients (Giannatos & Andrianopoulos, 1999).

2.3. Quantity of Road Transport in Europe

At a European level, freight transport has shown a remarkable increase in the last ten years. According to the European Commission's Directorate General for Mobility and Transport, the number of billions of tonne-kilometers travelled in 1995 was 2,846, rising to 3,661 billion in 2016 (+ 28%). Of these, 49% were travelled by road, 32% through shipping (within the EU), 11% through trains, 4% through inland maritime transport, 3% via pipelines and 1% via air.

The quantity of goods transported by road in the EU countries is escalating based on official Eurostat statistics. In terms of tonne-kilometres (tkm),¹ European road freight transport (RFT) increased by 4.5% in 2016 compared with 2015, with the 2016 value being the highest since 2012 (Eurostat, 2017). The total recorded quantity of million tkm on 2016 was 1,847,555. The largest proportion of tkm travelled was for national transport (64%), followed by international transport (23%), cross trade² (9%) and cabotage³ (2%). The top 5 countries in road freight transport were Germany, Poland, Spain, United Kingdom and France, and accounted for 62% of the total tkm travelled in the European Union. The largest increase between 2015 and 2016 in total tkm was recorded in Cyprus (+24.9%), followed by Romania (+23.5%) and Ireland (17.3%) (Eurostat, 2017).

According to Eurostat, in the EU28 for 2013, there were 563,598 road freight transport companies and most of them were met in Spain (19%), Poland (13%) and Italy (12%). In Europe 2.93 million people were employed in road transport sector. Some of the reasons for this large growth of the RFT

sector are summarized in the effort by companies to reduce costs (based on the rational application of logistics methods and techniques), to the need of combined road transport and to the fact that it is the most economical and flexible type of freight transport. Last but not least, the preference for this type of transport is mainly due to the fact that the road haulage market is also very competitive and fragmented, with a few large logistics companies and many small transport companies of three or fewer trucks (Harrison & Hoek, 2012).

2.4. Cost Comparisons

RFT companies incur a large number of costs in order to provide their services. Leaving aside the initial investment that is required for setting up a road transport company, the fixed and variable operating costs that are incurred for operating it are numerous. These may include the cost of fuels, tyres, maintenance, repairs, drivers' wages, parking space, tolls, road taxes, insurance, administrative expenses, accounting, phone bills, water, electricity, taxes etc. These costs differ considerably between European countries. However, there is a relatively very small number of researches regarding RFT costs and some of them will be mentioned in the following lines.

Bina et al. (2014) developed a model for comparing unit costs of road and rail freight transport for a sample of selected European countries and found that combined transport (rail and road), under certain infrastructure conditions, can have similar cost advantages as road transport. The European Commission's Directorate-General for Energy and Transport on 1998 funded the project SOFTICE (Survey on Freight Transport Including Cost Comparison for Europe). The project's results showed that drivers' wages were the largest cost factor, followed by fuels, and these factors varied substantially between countries. Total tax costs also varied between countries, ranging from 10% to 25% of the total operating cost of long haulage trucks. For 100km distance, EU prices were between 3 and 8 times higher in Western than in Eastern European countries, but prices were closer for long distance freight. According to a more recent report by AECOM (2014) for the European Commission, the largest share transport cost corresponds to drivers' wages and the cost of fuel. Smaller costs include maintenance, depreciation, taxes, insurance etc. This study also found large cost differences between European countries. Finally, Liakopoulou (2016) in her work analysed and measured the costs of road transportation companies and proposed a platform for the permanent measurement of these costs at company level. Lastly it must be mentioned that the term road transport costs, includes, apart from the monetary, also the environmental costs. Road transport creates environmental impacts from noise and air pollution and they have a negative impact on society and the environment (Milan, 2007). However, the measurement of these costs is not in the goals of the present research.

The literature review has pointed out that road transport is one of the most important means of freight transport in Europe, that it's a market that is growing, and that the quantity of RFT differs significantly from country to country. Moreover, from previous researches there seem to be considerable differences in the cost of RFT between countries. Consequently, the goal of the current research is to investigate, record and compare the cost of RFT in different European countries in order to make comparisons about cost competitiveness between them. The analysis and the conclusions of this research can be useful for European transport policy designers, who are interested in harmonising commercial road transport procedures and costs across European Union countries.

3. RESEARCH METHODOLOGY

The methodology that was followed in order to answer the research question was the following: first, a sample of 20 countries with the largest quantity total of road freight transport measured in tkm was chosen, using Eurostat data. These countries are shown in Table 1.

The sum of tkm of RFT in these 20 countries represents 96.3% of the total klm tonnes that were travelled in Europe in 2016. For each of these countries information was collected regarding 4 main

Table 1. Sample of European countries with the largest quantity of RFT

No.	Country	Total Million tkm
1	Germany	315,711
2	Poland	290,749
3	Spain	216,997
4	United Kingdom	176,678
5	France	155,843
6	Italy	112,637
7	Netherlands	67,964
8	Czech Republic	50,315
9	Romania	48,176
10	Sweden	42,673
11	Hungary	40,002
12	Slovakia	36,139
13	Bulgaria	35,409
14	Portugal	34,877
15	Lithuania	30,974
16	Belgium	30,865
17	Finland	26,837
18	Austria	25,082
19	Norway	20,932
20	Greece	20,903

Source: Eurostat with processing by the authors

cost categories: 1) fuels, 2) driver wages, 3) tolls and other road taxes and 4) tyres. Each cost category was approximated using the following method:

- **Fuels:** In order to approximate the cost of fuel in each country, the fuel consumption of a truck per 100 klm of highway road was multiplied by the cost of a litre of diesel in each country. The truck type that was used to estimate fuel consumption was based both on information collected a) by asking professional drivers and b) on data about truck sales in Europe. The truck that was pointed out as a very popular truck type was an 18 tonne, 12,000 cc Mercedes Actros with Euro 5 technology and a fuel consumption of 26.5 litre per 100 klm. The information about diesel litre cost was collected from the internet;
- **Driver cost:** The driver's labour cost was approximated by multiplying the hours that would be required by a professional truck driver to drive 100 klm of highway road at a speed of 80 klm per hour, by the labour cost per hour. However, from early on in the research it proved very difficult to collect information from companies and freelance drivers about the exact labour hour cost (salary + social insurance contributions) of a truck driver in each country. As a result, this cost was approximated by using the minimum wage per hour in each country. The minimum wage per hour was used under the assumption that a higher minimum wage means higher driver salaries, while a lower minimum wage means lower truck driver salaries. The minimum wage cost was

collected from Eurostat surveys and in cases of countries that don't use minimum wages the cost was approximated by the typical labour hour cost of an unskilled worker;

- **Tolls and other road taxes:** In order to approximate the cost of tolls, vignette and any other road taxes in each country, the authors estimated this cost for a 5-axle international truck travelling between each country's 2 biggest cities by population, and then apportioned this cost on a 100 klm scale;
- **Tyre cost:** The tyre cost was estimated by finding the tyre consumption of a 5-axle truck for a distance of 100 km. The estimations were made using the information that truck tyres typically last for about 120,000 klm and a common tyre type that is used for the Mercedes Actros is Michelin 315/70 R 22.5 X Line Energy D2. The authors calculated the cost of 10 tyres (5 axle truck) using prices from local Michelin dealers in each country.

4. ANALYSIS

4.1 Fuel Cost

The cost for the sample truck to travel 100 km of highway road in each of the 20 European countries is presented in Table 2. The columns present information about the fuel consumption of the sample truck, the price of diesel fuel per litre and the cost in euros (fuel consumption multiplied by the price of a litre of diesel).

In Table 2 considerable variations between countries can be observed. The lowest cost for fuel consumption is noted in Lithuania (26.77€), followed by Slovakia (27.30€) and the highest one in Norway (43.20€), followed by Italy (41.08€). The mean value for the 20 European countries was 34.85€ and 9 countries had values below average and 11 above average. The sample's variance was 20.98 with a standard deviation of 4.58.

4.2. Driver Cost

As mentioned in the methodology, the labour cost was approximated by multiplying the hours that would be required by a professional truck driver to drive 100 km of highway road at a speed of 80 km/hour, by the minimum wage per hour in each country. Table 3 presents the results.

From Table 3 it can be observed that the lowest labour cost is noted in Bulgaria (2.04€) followed by Lithuania (2.45€) and the highest one in Norway (18.66€), followed by Sweden (12.61€). The mean value for the 20 European countries was 8.07€ and 8 countries had values below average and 12 above average. The sample's variance was 29.47 and the standard deviation 5.45.

4.3. Tolls and Other Road Taxes

In order to approximate the cost of tolls, vignette and any other road taxes in each country, the authors made estimates about a 5-axle international truck travelling between each country's 2 biggest cities by population, and then apportioned this cost on a 100klm scale. The data are presented in Table 4.

From Table 4 it can be observed that the lowest cost is noted in Sweden that doesn't have any tolls, followed by Spain with 5.26€ and the highest one in Austria (41.00€) followed by Norway (30.07€). The mean value for the 20 European countries was 13.34€ and 12 countries had values below average and 8 above average. The sample's variance was 87.38 and the standard deviation 9.35.

4.4. Tyres Cost

As mentioned in the methodology, the tyre cost was estimated by finding the tyre consumption of a 5-axle truck for a distance of 100 km. The estimations were made using a tyre type that is common for the Mercedes Actros, the Michelin 315/70 R 22.5 X Line Energy D2. The authors calculated the cost of 10 tyres (5-axle truck) using prices from local Michelin dealers in each country as provided by the official Michelin European website (Table 5).

Table 2. Fuel cost in 20 European countries

No.	Country	Fuel Consumption in Diesel Litres per 100 km	Price of Litre of Diesel Fuel (20/7/2018)	Cost in Euros
1	Germany	26.5	1.30	34.45
2	Poland	26.5	1.13	29.95
3	Spain	26.5	1.23	32.60
4	United Kingdom	26.5	1.46	38.69
5	France	26.5	1.45	38.43
6	Italy	26.5	1.55	41.08
7	Netherlands	26.5	1.44	38.16
8	Czech Republic	26.5	1.26	33.39
9	Romania	26.5	1.20	31.80
10	Sweden	26.5	1.25	33.13
11	Hungary	26.5	1.24	32.86
12	Slovakia	26.5	1.03	27.30
13	Bulgaria	26.5	1.17	31.01
14	Portugal	26.5	1.43	37.90
15	Lithuania	26.5	1.01	26.77
16	Belgium	26.5	1.51	40.02
17	Finland	26.5	1.40	37.10
18	Austria	26.5	1.17	31.01
19	Norway	26.5	1.63	43.20
20	Greece	26.5	1.44	38.16

Notes: a) the fuel consumption of a 12.000 cc Mercedes Actros with Euro 5 technology was estimated at 26,5 litre per 100 km (<https://www.mercedes-benz.com/en/mercedes-benz/vehicles/trucks/fuel-comparison-tests-in-europe>) b) the price of diesel fuel was taken from the webpage <http://www.fuel-prices-europe.info/> for the data 20 July 2018 and were verified by the webpage <https://www.globalpetrolprices.com>. c) The exchange rate conversions to euros were made from the webpage <https://www.xe.com/> at the rate of 20/7/2018.

The lowest tyre cost is noted in Bulgaria with 4.55€, and the highest one in Norway (6.00€) followed by Finland with 5.99€. The mean value for the 20 European countries was 5.39€ and half of the countries had values below average and the other half above average. The sample's variance was 0.28 and the standard deviation 0.53.

4.5. Variance Analysis

Table 6 presents the total cost of RFT in the sample of 20 European countries.

From Table 6 it can be noticed that the lower road freight transport cost is met in Lithuania with 45.39€ followed by Poland with 45.85€ and Bulgaria with 48.59€. The highest cost is met in Norway with 102.59€, followed by Austria with 88.44€ and the UK with 74.90€. Also, the most important cost category is fuels, keeping in mind, however, that the driver wage cost is an approximation. Table 7 presents basic descriptive statistics about the costs and a measurement of standard deviation.

From Table 7 it can be noticed that the largest deviation is noted in the tolls and other road taxes cost, while the smallest one in the tyre cost. The total cost standard deviation from the average value of 61.56€ is 14.67€ and the width from the highest to the lowest total cost is 57.20€.

Table 3. Approximation of driver's labour cost in 20 European countries

No.	Country	Hours Needed to Travel 100 km in a Highway With a Speed 80 km/hour	Minimum Wage per Hour in Euros	Cost in Euros
1	Germany	1.25	8.84	11.05
2	Poland	1.25	3.17	3.96
3	Spain	1.25	4.94	6.18
4	UK	1.25	8.86	11.08
5	France	1.25	9.88	12.35
6	Italy	1.25	7.11	8.89
7	Netherlands	1.25	9.03	11.29
8	Czech Rep.	1.25	2.82	3.53
9	Romania	1.25	2.44	3.05
10	Sweden	1.25	11.32	14.15
11	Hungary	1.25	2.27	2.84
12	Slovakia	1.25	2.50	3.13
13	Bulgaria	1.25	1.63	2.04
14	Portugal	1.25	3.94	4.93
15	Lithuania	1.25	2.45	3.06
16	Belgium	1.25	9.49	11.86
17	Finland	1.25	5.55	6.94
18	Austria	1.25	8.92	11.15
19	Norway	1.25	18.66	23.33
20	Greece	1.25	3.94	4.93

Notes: a) the source from the minimum wage value was the report Statutory minimum wages 2018 by Eurofound (https://www.eurofound.europa.eu/sites/default/files/ef_publication/field_ef_document/ef18005en.pdf). b) Italy, Sweden, Finland, Austria and Norway do not have minimum wages. For Italy the minimum wage per hour was calculated by the monthly unemployment benefit (1,195 euros) divided by the typical workload of 168 working hours per month (4.2 weeks*40 hours) (Source: <https://www.eurofound.europa.eu>). For Sweden it was calculated from the salary of a hotel employee aged 20 and above, with no experience and no education (<https://www.quora.com/Whats-the-minimum-wage-in-Sweden>). For Finland the minimum wage must be at least 40 times the basic daily unemployment assistance. Since 1 January 2007 this has meant that the paid wages should be at least €956.10 per month, €44.47 per day or €5.55 per hour (Source: <https://www.eurofound.europa.eu/publications/report/2009/finland-wage-formation>). In Austria the minimum wage has been voted and will be applied in 2020. Its height will be 1,500€ per month (Source: <https://www.eurofound.europa.eu>). In Norway trade unions in each sector bargain different rates. The rate used was taken from unexperienced worked in construction industry (177 krone per hour) (Source: <https://www.lifeinnorway.net>).

5. CONCLUSION

As pointed out by the literature review, road transport is one of the most important means of freight transport globally, and the quantity of RFT differs significantly from country to country. The goal of the current research was to investigate, record and compare the cost of RFT in various European countries in order to make comparisons between them. As mentioned before, the analysis and the conclusions of this research can be useful for European policy designers, who wish to harmonize commercial road transport procedures and costs across European Union countries.

The results of the current research have shown that there are considerable differences in the RFT cost between the 20 European countries of the sample. The countries found to have the lowest RFT cost were Lithuania, Poland and Bulgaria, while the ones with the highest costs were Norway,

Table 4. Calculations of tolls and other road taxes in 20 European countries

No.	Country	1st Largest City	2nd Largest City	Distance in km	Cost in Euros	Reason	Cost for 100 km in Euros
1	Germany	Berlin	Hamburg	287	44.77	Tolls	15.60
2	Poland	Warsaw	Krakow	360	22.57	Tolls	6.27
3	Spain	Madrid	Barcelona	617	32.45	Tolls	5.26
4	UK	London	Birmingham	126	24.30	Levy cost and tolls	19.29
5	France	Paris	Marseille	773	59.50	Tolls	7.70
6	Italy	Rome	Milan	572	101.40	Tolls	17.73
7	Netherlands	Amsterdam	Rotterdam	95	8.00	Eurovignette	8.00
8	Czech Rep.	Prague	Brno	186	33.93	Tolls	18.24
9	Romania	Bucharest	Iasi	389	11.00	Vignette	11.00
10	Sweden	Stockholm	Goetnberg	468	8.00	Eurovignette	8.00
11	Hungary	Budapest	Debrecen	230	8.00	Eurovignette	8.00
12	Slovakia	Bratislava	Košice	397	73.50	Tolls	18.51
13	Bulgaria	Sofia	Plovdiv	133	11.00	Vignette	11.00
14	Portugal	Lisbon	Porto	312	22.55	Tolls	7.23
15	Lithuania	Vinius	Kaunas	92	11.00	Vignette	11.00
16	Belgium	Antwerp	Ghent	59	7.39	Tolls	7.39
17	Finland	Helsinki	Tampere	178	0.00	No tolls	0.00
18	Austria	Vienna	Graz	195	79.95	Tolls	41.00
19	Norway	Oslo	Bergen	305	91.71	Tolls	30.07
20	Greece	Athens	Thessaloniki	499	77.65	Tolls	15.56

Notes: The information was drawn from the following sources: 1. Germany Toll Collect <https://www.toll-collect.de/en>, 2. Poland <https://www.tolls.eu>, <https://www.multiservicetolls.com/>, 3. Spain <https://www.viamichelin.com>, 4. UK <https://www.gov.uk/uk-toll-roads>, <https://www.multiservicetolls.com>, 5. France <http://www.autoroutes.fr>, 6. Italy <https://www.autostrade.it>, 7. Netherlands <https://www.dkv-euroservice.com>, 8. Czech Rep. <http://www.myto.cz.eu>, 9. Romania <https://www.untr.ro>, <https://www.tolls.eu>, 10. Sweden <https://www.dkv-euroservice.com>, 11. Hungary <https://www.hu-go.hu/>, 12. Slovakia <https://www.dkv-euroservice.com>, <https://www.emyto.sk>, 13. Bulgaria <https://www.tolls.eu>, 14. Portugal <http://www.portugal.tolls.com/>, 15. Lithuania <https://lakd.lrv.lt/en/road-charges-and-tolls/user-charge-vignettes>, 16. Belgium <https://www.viapass.be>, 17. Finland <http://www.highwaymaps.eu/finland>, 18. Austria, <https://www.asfinag.at>, 19. Norway <https://www.fjellinjen.no/private/toll-calculator/>, 20. Greece <http://diodia.com.gr/>.

Austria and UK. The largest differences in costs were met in tolls and other road taxes, followed by drivers' wages, fuels and tyres. Harmonization policies for RFT among EU countries may include the application of common policies regarding the cost of tolls as well as the cost of fuels, which proved to be the most important cost categories. The high standard deviation in the cost of tolls may signal the possibility of designing a common RFT policy across all EU countries, while the cost of fuels may be harmonized with setting a common tax rate among them. Another proposal would be to promote the development of "greener" commercial tracks, which can be fueled by electricity produced from renewable energy sources across all EU countries.

It is necessary to mention though, that the current research has limitations. Two of the most important ones are related with the fact that there are various other cost categories that are not measured in the research, as well as the lack of data on actual drivers' wages, which were found to be the most important cost category in other researches. Further research could shed light on these cost categories, as well as expand the sample on more counties both within and outside the European Union.

Table 5. Tyre consumption in euros for 100 km of highway road for a 5 axle Mercedes Actros

No.	Country	Cost of 1 Tyre	Cost of 10 Tyres	Tyre Duration in km	Cost for 100 km in Euros
1	Germany	634.2	6,342	120,000	5.29
2	Poland	680.92	6,809.2	120,000	5.67
3	Spain	708.99	7,089.9	120,000	5.91
4	UK	702	7,020	120,000	5.85
5	France	697.2	6,972	120,000	5.81
6	Italy	693.88	6,938.8	120,000	5.78
7	Netherlands	636.4	6,364	120,000	5.30
8	Czech Rep.	621	6,210	120,000	5.18
9	Romania	558.27	5,582.7	120,000	4.65
10	Sweden	718.37	7,183.7	120,000	5.99
11	Hungary	650	6,500	120,000	5.42
12	Slovakia	573	5,730	120,000	4.78
13	Bulgaria	545.55	5,455.5	120,000	4.55
14	Portugal	708.99	7,089.9	120,000	5.91
15	Lithuania	547.16	5,471.6	120,000	4.56
16	Belgium	636.4	6,364	120,000	5.30
17	Finland	718.37	7,183.7	120,000	5.99
18	Austria	634	6,340	120,000	5.28
19	Norway	720.33	7,203.3	120,000	6.00
20	Greece	550	5,500	120,000	4.58

Table 6. Totals of RFT cost for 100 km of highway road for 20 European countries

No.	Country	Fuel Cost	Driver Cost	Tolls	Tyre Cost	Total Cost in Euros
1	Lithuania	26.77	3.06	11.00	4.56	45.39
2	Poland	29.95	3.96	6.27	5.67	45.85
3	Bulgaria	31.01	2.04	11.00	4.55	48.59
4	Hungary	32.86	2.84	8.00	5.42	49.11
5	Spain	32.60	6.18	5.26	5.91	49.94
6	Finland	37.10	6.94	0.00	5.99	50.02
7	Romania	31.80	3.05	11.00	4.65	50.50
8	Slovakia	27.30	3.13	18.51	4.78	53.71
9	Portugal	37.90	4.93	7.23	5.91	55.96
10	Czech Rep.	33.39	3.53	18.24	5.18	60.33
11	Netherlands	38.16	11.29	8.00	5.30	62.75
12	Sweden	33.13	14.15	8.00	5.99	61.26
13	Greece	38.16	4.93	15.56	4.58	63.23
14	France	38.43	12.35	7.70	5.81	64.28
15	Belgium	40.02	11.86	7.39	5.30	64.57
16	Germany	34.45	11.05	15.60	5.29	66.38
17	Italy	41.08	8.89	17.73	5.78	73.47
18	UK	38.69	11.08	19.29	5.85	74.90
19	Austria	31.01	11.15	41.00	5.28	88.44
20	Norway	43.20	23.33	30.07	6.00	102.59

Table 7. Descriptive statistics and variance analysis

No.	Fuel Cost	Driver Cost	Tolls	Tyre Cost	Total Cost
Min	26.77	2.04	0.00	4.55	45.39
Max	43.20	23.33	41.00	6.00	102.59
Width	16.43	21.29	41.00	1.45	57.20
Average	34.85	7.99	13.34	5.39	61.56
Standard deviation	4.58	5.34	9.35	0.53	14.67

REFERENCES

- AECOM. (2014). *Report on the State of the EU Road Haulage Market*. European Commission, Directorate General for Mobility and Transport.
- Banister, D., & Berechman, J. (1999). *Transport investment and economic development*. Taylor and Francis.
- Bína, L., Bínová, H., Březina, E., Kumpošt, P., & Padělek, T. (2014). Comparative model of unit costs of road and rail freight transport for selected European countries. *European Journal of Business and Social Sciences*, 3(4), 127–136.
- Branch, A. (2009). *Global supply chain management and international logistics*. Routledge.
- Britannica. (2018). *Wheel*. Retrieved from <https://www.britannica.com/technology/wheel>
- Dell, R., Moseley, P., & Rand, D. (2014). *Towards Sustainable Road Transport*. Academic Press.
- Erb, J., & James, S. (2017). *Global logistics for dummies*. John Wiley & Sons.
- European Commission, Directorate-General for Mobility and Transport in co-operation with Eurostat. (2019). *Transport in Figures 2018*.
- Eurostat. (2017). *Road freight transport statistics*. Retrieved from https://ec.europa.eu/eurostat/statistics-explained/index.php/Road_freight_transport_statistics
- Freitag, D. (1979). History of wheels for off-road transport. *Journal of Terramechanics*, 16(2), 49–68. doi:10.1016/0022-4898(79)90001-6
- Giannatos, G., & Andrianopoulos, S. (1999). *Logistics, Transport and Distribution*. Tekdotiki.
- Grant, D., Trautrim, A., & Yew Wong, C. (2015). *Sustainable Logistics and Supply Chain Management*, Kogan Page Limited.
- Harrison, A., & Hoek, R. (2012). *Logistics Management & Strategy*. Rosili.
- Liakopoulou, S. (2016). *Cost and pricing in road freight transport: Development of a platform for calculating the cost of transport*. Aristotle University of Thessaloniki Polytechnic School.
- Milan, J. (2007). Modelling the full cost of an intermodal and road freight transport network. *Transportation Research Part D, Transport and Environment*, 12(1), 33–44. doi:10.1016/j.trd.2006.10.004
- Spiekermann, M., & Urban, W. (2013). *Overview of Country Transport Models*. Dortmund: Regional Section Research Lindemannstrasse.
- Theodoropoulou, R., & Kasoli, M. (2014). *Transport and Logistics*. Hellenic General Secretariat for Research and Technology.

ENDNOTES

- ¹ A tonne-kilometre, abbreviated as tkm, is a unit of measure of freight transport which represents the transport of one tonne of goods (including packaging and tare weights of intermodal transport units) by a given transport mode (road, rail, air, sea, inland waterways, pipeline etc.) over a distance of one kilometre.
- ² Cross-trade is international road transport between two different countries performed by a road motor vehicle registered in a third country.
- ³ Cabotage is road transport by a motor vehicle registered in a country performed on the national territory of another country. Cabotage data are reported by European Union (EU) Member States for hauliers registered in their country.

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